(iii) $0.862 = \frac{862}{1000}$

$= \frac{862 \div 2}{1000 \div 2} = \frac{431}{500}$

Example 1.10 Find the decimal form of the following fractions.

(i) $153 + 96 + 7 + \frac{5}{10} + \frac{2}{1000}$ (ii) $999 + 99 + 9 + \frac{9}{10} + \frac{9}{100}$ (iii) $23 + \frac{6}{10} + \frac{8}{1000}$

Solution

(i) $153 + 96 + 7 + \frac{5}{10} + \frac{2}{1000} = 256 + 5 \times \frac{1}{10} + 0 \times \frac{1}{100} + 2 \times \frac{1}{1000}$

$= 256.502$ (since hundredths place is not there, it is taken as '0')

(ii) $999 + 99 + 9 + \frac{9}{10} + \frac{9}{100} = 1107 + 9 \times \frac{1}{10} + 9 \times \frac{1}{100}$

$= 1107.99$

(iii) $23 + \frac{6}{10} + \frac{8}{1000} = 23 + 6 \times \frac{1}{10} + 0 \times \frac{1}{100} + 8 \times \frac{1}{1000}$

$= 23.608$ (since hundredths place is not there, it is taken as '0')

Example 1.11 Write each of the following as decimals.

(i) Four hundred four and five hundredths

(ii) Two and twenty five thousandths.

Solution

(i) Four hundred four and five hundredths.

$= 404 + \frac{5}{100}$

$= 404 + 0 \times \frac{1}{10} + 5 \times \frac{1}{100} = 404.05$

(ii) Two and twenty five thousandths

$= 2 + \frac{25}{1000}$

$= 2 + \frac{2}{100} + \frac{5}{1000}$ $\left[\text{since, } \frac{25}{1000} = \frac{20+5}{1000} = \frac{20}{1000} + \frac{5}{1000} = \frac{2}{100} + \frac{5}{1000}\right]$

$= 2 + \frac{0}{10} + \frac{2}{100} + \frac{5}{1000} = 2.025$ [as there is no tenth we take it as 0 tenth]

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